MHF4U Advanced Functions - Unit 1 Practice Test, Version 5

Worked key and thinking guide

60 marks: 10 multiple choice + 50 multipart. Award equivalent correct reasoning. Graphs require labelled key points and correct open or closed endpoints.

Part A - Multiple choice [10]

Question	Answer	Reason
1	A	Every input has one output, but inputs 0 and 1 share output 1.
2	C	Require `2x-6>=0` and exclude `x=5`, where the denominator is zero.
3	B	The downward V has maximum 5.
4	A	`-2(x-4)=9` gives `x=-1/2`; `-3(3)+2=-7`.
5	D	Restriction `x>=1` selects the positive square-root branch.
6	B	`(-x)^3-4(-x)=-(x^3-4x)`.
7	C	Left-hand value is `3(2)-1=5`; `2k+1=5`, so `k=2`.
8	A	`f(1)=f(3)=-3`; the secant slope is zero.
9	D	The centred slope is 2; the two endpoint heights differ by 0.4 over 0.2.
10	C	Negative secant means `f(b)<f(a)`; zero tangent means locally flat at a.

Part B - Multipart questions [50]

11. Piecewise functions [9]

a. First branch is a downward parabola from filled `(-3,0)` through vertex `(-1,4)` towards open `(0,3)`. Second branch is a line from filled `(0,3)` to filled `(2,7)`. Third branch is a decreasing line from open `(2,5)` to filled `(5,-1)`. The open `(0,3)` of the first branch is covered by the second branch's filled point. [4]

b. Domain `[-3,5]`; range `[-1,7]`. [2]

c. Increasing on `[-3,-1]` and `[0,2]`. [1]

d. No. `F(-1)=4` and `F(1/2)=4`, so an inverse relation would assign input 4 two different outputs. [2]

Thinking move: Check each branch's endpoint, then combine its output values to find the full range.

12. Absolute value graph [6]

The vertex is `(-1,6)` and the V opens downward. `-3|x+1|+6=0` gives `|x+1|=2`, so x-intercepts are `(-3,0)` and `(1,0)`. The y-intercept is `(0,3)`. Domain `(-infinity,infinity)`; range `(-infinity,6]`. The negative sign reflects the parent graph in the x-axis, and 3 stretches it vertically by factor 3. [3 graph; 2 domain/range; 1 explanation]

Thinking move: Check the y-intercept separately by substituting `x=0`; it is not necessarily a root.

13. Transforming a function [5]

a. Match the original input 9: `-2(x-4)=9`, so `x=-1/2`. The new output is `-3f(9)+2=-3(3)+2=-7`; point `(-1/2,-7)`. [2]

b. Require `-2(x-4)>=0`, giving domain `(-infinity,4]`. Since `f` is nonnegative and multiplied by `-3`, range `(-infinity,2]`. [2]

c. Reflect horizontally in the y-axis, compress horizontally by factor `1/2`, and shift right 4. The complete point map is `(u,v)->(4-u/2,-3v+2)`. [1]

Thinking move: The inside negative reverses the domain direction; the outside negative reverses the range direction.

14. Inverse relations [5]

a. Original q has domain all real numbers and range (-infinity,7]. Its inverse relation has domain (-infinity,7] and range (-infinity,infinity). [2]

b. The parabola fails the horizontal-line test: q(1)=5=q(3), so the inverse would send input 5 to two outputs. [1]

c. One valid restriction is original x>=2. From y=-2(x-2)^2+7, rearrange to (x-2)^2=(7-y)/2. Then q^(-1)(x)=2+sqrt((7-x)/2), with domain (-infinity,7]. Also accept original x<=2 with inverse 2-sqrt((7-x)/2) and the same inverse domain. [2]

Thinking move: Restrict to one side of the vertex before choosing the square-root sign.

15. Circular ride diagram [7]

a. Centre height 1.5+4.5=6 m; greatest height 6+4.5=10.5 m. [2]

b. Domain [0,420] s; range [1.5,10.5] m. [2]

c. Six metres is the centre height, not the top. Add one more radius of 4.5 m to get 10.5 m. [1]

d. Draw a smooth cycle through (0,1.5), (105,6), (210,10.5), (315,6), and (420,1.5). Label time in seconds and height in metres. [2]

Thinking move: Divide the rotation time by four to locate the quarter-turn heights.

16. Average and instantaneous rates [8]

a. H(1)=-1+10+4=13 and H(3)=-9+30+4=25; average rate (25-13)/(3-1)=6 m/s. [2]

b. H(2.9)=24.59 and H(3.1)=25.39; centred estimate (25.39-24.59)/(3.1-2.9)=4 m/s. [3]

c. Complete the square: H(t)=-(t-5)^2+29, so the maximum modelled height is 29 m at t=5 s. [2]

d. Part a averages the rate from 1 to 3 seconds; part b estimates the rate close to 3 seconds. The upward rate is changing. [1]

Thinking move: An interval's average rate can differ from the rate at its right endpoint.

17. Graph properties and interval notation [5]

a. r(-x)=(-x)^4-3(-x)^2+2=x^4-3x^2+2=r(x), so r is even. [2]

b. Y-axis symmetry gives (-2,6). [1]

c. |2x+1|<5 means -5<2x+1<5, so -6<2x<4, -3<x<2, or (-3,2). [2]

18. Secant and tangent diagram [5]

a. Secant AB gives the average rate on [2,4]: (2-6)/(4-2)=-2 units of y per unit of x. [2]

b. The horizontal tangent at A gives instantaneous rate at x=2: slope 0. [2]

c. From A to B the height of the curve falls overall, giving a negative secant slope. At its maximum A, the curve is locally flat, giving tangent slope zero. [1]

Thinking move: A secant describes change across an interval; a tangent describes direction at one x-value.